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 Studierichting: Informatica
 Oefeningenreeks 5B nr. 5.10

$$\int_0^{\frac{\pi}{2}} \frac{\sin(x)dx}{\sin(x) + \cos(x)}$$

Primitieve vinden:

$$t = \tan\left(\frac{x}{2}\right)$$

$$dx = \frac{2}{1+t^2}dt$$

$$\begin{aligned} & \int \frac{\frac{2t}{1+t^2}}{\frac{-t^2+2t+1}{1+t^2}} \frac{2dt}{1+t^2} \\ &= \int \frac{4t}{(-t^2+2t+1)(1+t^2)} dt \\ &= \int -\frac{4t}{(t-(1+\sqrt{2}))(t-(1-\sqrt{2}))(1+t^2)} dt \end{aligned}$$

Stelling Fuss (3)

$$-\frac{4t}{(t-(1+\sqrt{2}))(t-(1-\sqrt{2}))(1+t^2)} = \frac{A}{t-(1+\sqrt{2})} + \frac{B}{t-(1-\sqrt{2})} + \frac{Ct+D}{(1+t^2)}$$

Voor $t = 1 + \sqrt{2}$:

$$A = \frac{-4t}{(t-(1-\sqrt{2}))(1+t^2)} = \frac{-4(1+\sqrt{2})}{(2\sqrt{2})(4+2\sqrt{2})} = \frac{-4(1+\sqrt{2})}{(8\sqrt{2}+8)} = \frac{-4}{8} = -\frac{1}{2}$$

Voor $t = 1 - \sqrt{2}$:

$$\begin{aligned} B &= \frac{-4t}{(t-(1+\sqrt{2}))(1+t^2)} = \frac{-4(1-\sqrt{2})}{(-2\sqrt{2})(4-2\sqrt{2})} = \frac{-4(1-\sqrt{2})}{(-8\sqrt{2}+8)} = -\frac{4}{8} \\ &= -\frac{1}{2} \end{aligned}$$

Voor $t = i$:

$$\begin{aligned} Ci + D &= \frac{-4t}{(t-(1+\sqrt{2}))(t-(1-\sqrt{2}))} = \frac{-4i}{(i-(1+\sqrt{2}))(i-(1-\sqrt{2}))} \\ &= \frac{-4i}{(i-1-\sqrt{2})(i-1+\sqrt{2})} = \frac{-4i}{(-2i-2)} = \frac{2i}{i+1} \\ &= \frac{2i(i-1)}{(i+1)(i-1)} = \frac{-2-2i}{-2} = i+1 \end{aligned}$$

$(C, D) = (1, 1)$ dus $C = 1$ en $D = 1$

$$\frac{4t}{(-t^2 + 2t + 1)(1 + t^2)} = \frac{-1}{2(t - (1 + \sqrt{2}))} + \frac{-1}{2(t - (1 - \sqrt{2}))} + \frac{t + 1}{(1 + t^2)}$$

Integralen berkenen:

$$-\frac{1}{2} \int \frac{dt}{t - (1 + \sqrt{2})}$$

$$u = t - (1 + \sqrt{2})$$

$$du = dt$$

$$= -\frac{1}{2} \int \frac{du}{u} = -\frac{1}{2} \ln(u) + C = -\frac{1}{2} \ln |t - (1 + \sqrt{2})| + C$$

$$-\frac{1}{2} \int \frac{dt}{t - (1 - \sqrt{2})}$$

$$u = t - (1 - \sqrt{2})$$

$$du = dt$$

$$= -\frac{1}{2} \int \frac{du}{u} = -\frac{1}{2} \ln(u) + C = -\frac{1}{2} \ln |t - (1 - \sqrt{2})| + C$$

$$\int \frac{t + 1}{t^2 + 1} = \int \frac{t}{t^2 + 1} dt + \int \frac{1}{t^2 + 1} dt$$

$$u = t^2 + 1$$

$$du = 2t dt$$

$$= \frac{1}{2} \int \frac{du}{u} + \text{Bgtan } t = \frac{1}{2} \ln |u| + \text{Bgtan } t + C = \frac{1}{2} \ln |t^2 + 1| + \text{Bgtan } t + C$$

Samengesteld:

$$-\frac{1}{2} \ln |(t - (1 + \sqrt{2}))| - \frac{1}{2} \ln |t - (1 - \sqrt{2})| + \frac{1}{2} \ln |t^2 + 1| + \text{Bgtan } t + C$$

$$= \frac{1}{2} \ln \left| \frac{t^2 + 1}{(t - (1 + \sqrt{2}))(t - (1 - \sqrt{2}))} \right| + \text{Bgtan } t + C$$

$$t = \tan \left(\frac{x}{2} \right):$$

$$\text{We weten } \sin(x) + \cos(x) = \frac{-t^2 + 2t + 1}{1 + t^2} \text{ dus } \frac{t^2 + 1}{t^2 - 2t - 1} = -\frac{1}{\sin(x) + \cos(x)}:$$

$$\text{En } \text{Bgtan} \left(\tan \left(\frac{x}{2} \right) \right) = \frac{x}{2}:$$

$$\begin{aligned}
& \text{Dus } -\frac{1}{2} \ln |t - (1 + \sqrt{2})| - \frac{1}{2} \ln |t - (1 - \sqrt{2})| + \frac{1}{2} \ln |t^2 + 1| + \frac{x}{2} + C \\
&= \frac{1}{2} \ln \left| -\frac{1}{\sin(x) + \cos(x)} \right| + \text{Bgtan } t + C = \frac{1}{2} \ln \left| -\frac{\sin^2(x) + \cos^2(x)}{\sin(x) + \cos(x)} \right| + \frac{x}{2} + C \\
&= -\frac{1}{2} \ln |\sin(x) + \cos(x)| + \frac{x}{2} + C
\end{aligned}$$

Grenzen invullen:

$$-\frac{1}{2} \ln |1| + \frac{\pi}{4} + \frac{1}{2} \ln |1| - \frac{0}{2} = \frac{\pi}{4}$$

De integraal is convergent!

Opmerking WP: Dat is juist, maar er is een veel kortere methode die ik u niet wil onthouden. Stel

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin(x) dx}{\sin(x) + \cos(x)}$$

Stel $t = \frac{\pi}{2} - x$, dan is

$$I = - \int_{\frac{\pi}{2}}^0 \frac{\cos(t) dt}{\cos(t) + \sin(t)}$$

Tellen we de twee integralen op, dan is

$$2I = \int_0^{\frac{\pi}{2}} \frac{(\sin(x) + \cos(x)) dx}{\sin(x) + \cos(x)} = \int_0^{\frac{\pi}{2}} dx = \frac{\pi}{2}$$

dus $I = \frac{\pi}{4}$

References